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Palaeomagnetism of the (Early Devonian) Lower Old Red Sandstones of South Wales: implications to Variscan overprinting and differential regional rotations

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Abstract

Component structures of magnetic remanence in the Old Red Sandstones of South Wales are related to tectonic history by detailed study of three contrasting areas: (1) remote from, (2) bordering, and (3) within the Variscan orogenic belt. A uniform southerly shallow-directed component dominates remanence in each sector. In area 1, where it is considered most representative of the regional field, this magnetisation has a mean $D/I = 191.1/-8.0°$ (151 samples in situ, $\alpha_{95} = 2°$) equivalent to a pole position at 341.1°E, 41.2°S. Fold tests show that it was acquired at an intermediate stage of orogenesis (and during the Carboniferous–Permian Reversed Superchron). It is interpreted as a pervasive magnetic overprint imparted by northward fluid migration consequent upon loading of the orogen to the south. Remote from the orogenic front, a higher blocking temperature dual-polarity remanence of probable Lower Devonian age has sporadically survived ($D/I = 246/38°$, 27 samples tilt adjusted, $\alpha_{95} = 7.2°$) corresponding to a pole position at 297°E, 3°N. This remanence is entirely replaced close to the frontal zone of the orogen. Within the orogen, low blocking temperature components are more prominent and chemical alteration obscures the highest blocking temperature portion of the remanence spectrum. This region was deformed into upright folds before the magnetising event.

Following remanence acquisition, continuing deformation produced asymmetry in fold structure producing a plunge and steepening of the southern limbs. Structures in the south were already more deformed than those in the north at the time of magnetisation, implying that the causative fluid front moved northwards with time. The palaeomagnetic vectors are rotated clockwise to the west in South Wales and, in part, record the effects of regional dextral transpression; the largest rotation however, is observed within the same thrust block and apparent differential rotation here is probably a consequence of strain.

1. Introduction

The Lower Old Red Sandstones in South Wales have been of palaeomagnetic interest since the earliest investigations (Creer, 1957) and a multicomponent magnetisation structure was recognised at an early stage (Creer, 1962; Chamalaun, 1964; Chamalaun and Creer, 1964). More recent investigation (Channell et al., 1992) studied an Acadian (late Caledonian) structure to clarify the direction of the prefolding component ($D/I = 232/32°$). The most prominent part of the remanence spectrum however, is a component with a shallow southerly direction which

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appears to postdate Acadian structures in the north of the outcrop (Channell et al., 1992) but which itself becomes folded as the Variscan (late Carboniferous) fold belt is entered in the south (Stearns and Van der Voo, 1987).

The significance of remanence recorded by Old Red Sandstones within the Variscan fold belt is disputed. McClelland Brown (1983) recognised both syn-folding and pre-folding components which could only be reconciled with the Upper Carboniferous and Lower Devonian field directions by postulating a 40° clockwise rotation of the whole of southern Pembrokeshire. Stearns and Van der Voo (1987) however, found no rotation of the pre-folding component at a nearby location within the belt and considered that no post-Hercynian rotation had taken place here. The two studies might be reconciled if the structures sampled by Stearns and Van der Voo (1987) postdate a regional clockwise rotation (see McClelland, 1987).

The present regional investigation of the Old Red Sandstone is conducted in three contrasting tectonic settings. It employs deformed structures ranging from tight folds in the Variscan belt in the south to weakly deformed beds on the foreland platform in the north, to routinely apply the palaeomagnetic fold test (Graham, 1949). As a result, the progressive effects of Variscan overprinting and deformation can be evaluated. In addition, the areal extent of regional rotations in southwestern Wales at the northern border of the Variscan Orogeny is clarified.

2. Geological outline and sampling

In southwest Wales the WNW–ESE-trending Variscan (Hercynian) orogenic belt, deformed predominantly in the Late Carboniferous (Owen, 1974; Hancock et al., 1983), impinges on the NE–SW-trending paratectonic (non-metamorphic) Caledonides (Fig. 2) where deformation was concentrated in the Acadian (late Silurian–mid-Devonian) orogenic episode. A cuspate sedimentary basin (the Anglo-Welsh Cuvette, see George, 1970) comprises a thick wedge of predominantly

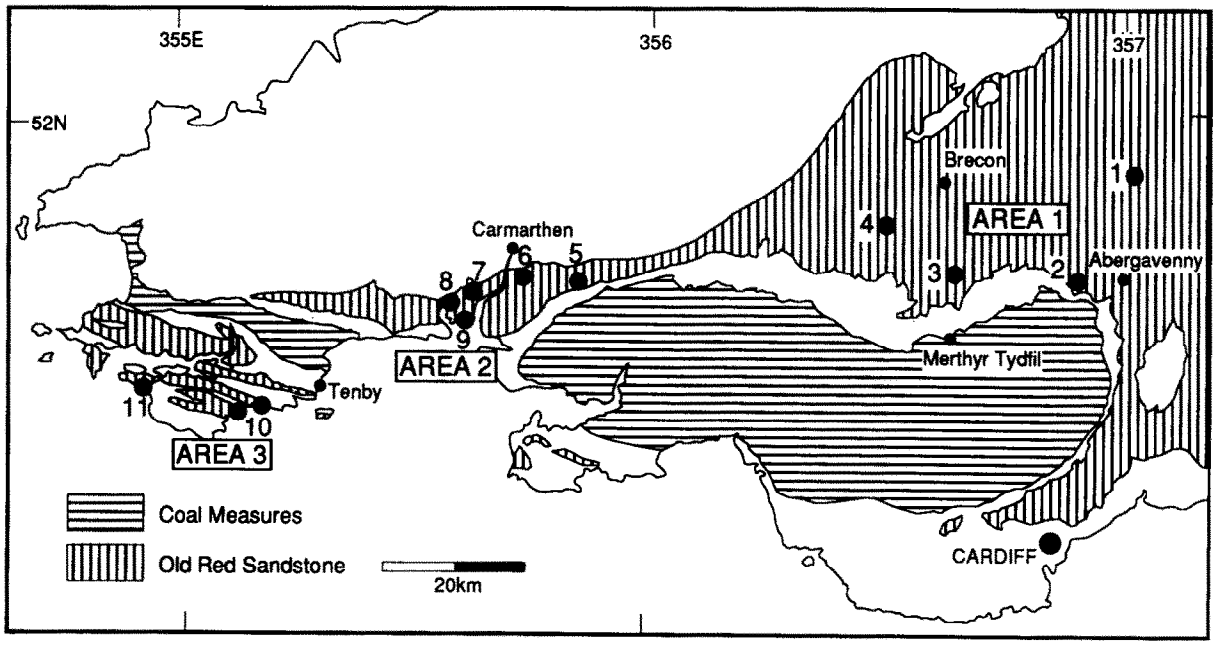


Fig. 1. Outline geological map of South Wales illustrating outcrops of the Old Red Sandstone and Coal Measures formations and distribution of sampling locations of this study. Each sampling location comprises a number of distributed sites as summarised in Table 1.

fluviatile sediments deposited as detritus eroded from the uprising Acadian Orogeny to the northwest. Formerly assigned to the Lower Old Red Sandstone, deposition spanned Pridolian (latest Silurian) to Emsian times (House et al., 1977). Numerous fining-upwards sequences are interpreted as braided stream deposits (Allen, 1974). An unconformity formed during mid-Devonian times defines uplift and erosion associated with Acadian deformation.

Subsequent deformation during the Variscan episode increased in intensity to the south and the outcrop can be zoned on the basis of structural style (Hancock et al., 1981). The outer (northern) Zone II (Fig. 2b) contains open folds with a WSW trend. The succeeding Zone Ic has small, non-cylindrical folds with abundant décollement surfaces. Zone Ib has larger folds which plunge more steeply and are complicated by parasitic folds. Interior Zone Ia is characterised by gently plunging folds of large amplitude. This tectonic progression is linked to two major thrusts (Fig. 2b). The Ritec–Tenby thrust at the base of Zone Ia appears to have migrated northwards

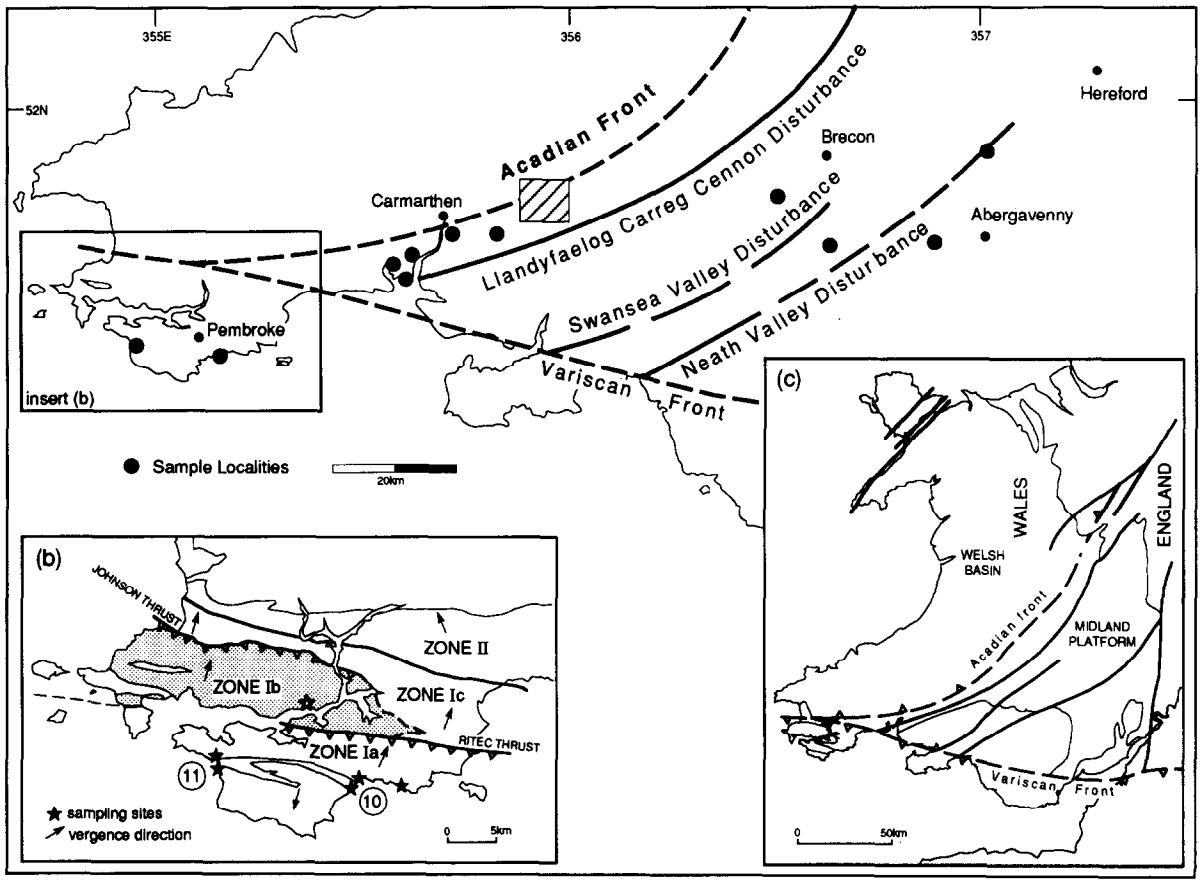


Fig. 2. Distribution of palaeomagnetic sampling sites in the Old Red Sandstones of South Wales in relationship to the major tectonic features. The shaded boxed area is the study location of Channell et al. (1992) spanning an Acadian structure. (b) The southwestern part of the study area illustrating tectonic domains noted in the text (see Hancock et al., 1983) and sample locations 10 and 11 of this study. The open star is the sampling location of Stearns and Van der Voo (1987); the sampling sites of McClelland Brown (1983) are at the same locations as sites of this study. (c) Regional tectonic setting of the Old Red Sandstone outcrop in South Wales illustrating the Acadian and Variscan orogenic fronts (dashed lines) and major fault zones (solid lines).

Table 1
Numbers and locations of the palaeomagnetic sampling sites in Lower Old Red Sandstones of South Wales

Site No.	Dip/dir.	No. cores
Area 1, Brecon and Abergavenny District		
Locality 1 (ca. SO2860 2530), Llanthony		
1	40/140	13
2	40/140	8
3	2/275	20
Locality 2 (ca. SO2320 1350), Clydach		
4	15/165	5
5	15/177	5
6	17/136	5
7	19/160	5
8	15/165	5
9	15/165	5
10	15/165	5
11	15/165	5
12	15/165	5
13	15/165	5
14	15/165	4
Locality 3 (ca. SO0630 1230), Pontiscill		
15	4/130	13
16	4/130	6
Locality 4 (ca. SN9715 2080), Storey Arms		
17	6/130	7
18	6/130	12
19	6/130	9
Area 2, Carmarthen District		
Locality 5 (ca. SN5150 1660), Llanddarog		
20	57/168	6
21	62/162	10
22	62/162	10
Locality 6 (ca. SN4245 1620), Oaksland		
23	19/132	11
24	24/123	9
25	14/140	7
Locality 7 (ca. SN3490 1465), Pen-y-coed		
26	38/192	5
27	38/192	9
28	38/170	15
Locality 8 (ca. SN3145 1245), Pilgrim Rest		
29	39/172	19
30	39/172	6
Locality 9 (ca. SN3450 0950), Llanstephan		
31	18/167	5
32	10/190	5
33	48/168	5
34	40/340	5
35	64/333	3
36	45/349	5
37	45/349	5
38	65/342	5
39	65/342	5
40	54/330	15
41	15/166	18
42	14/180	3
43	0/0	7
44	4/251	8
45	0/0	7
46	0/0	7
47	41/327	5
48	32/324	9
Area 3, southwest Pembrokeshire		
Locality 10 (ca. SS0220 9780), Freshwater East		
49 [a]	90/0	24
50	82/6	19
51	42/184	13
Locality 11 (ca. SR8840 9950), Freshwater West		
52	96/184	18
53	52/22	16

Locations in brackets are National Grid reference numbers. See Figs. 1 and 2 for general locations.
[a] This site includes samples from Manorbrier Bay (SS0550 9769).

and propagated westwards before becoming pinned and producing a fan of folds on the hanging-wall side (M.P. Coward, in McClelland Brown, 1983). Thrusting then migrated northwards to form the Johnston thrust which underlies Zone Ib; this thrust propagated eastwards where it eventually died out in fold structures. The net result of thrusting was to produce a fan-shaped complex of folds, becoming more south-facing towards the south. Zones Ic and II north of the thrust are dominated by thin-skinned tectonics.

Sample localities in area 1 (localities 1–4, Fig. 1) are remote fom the Variscan front and only mildly deformed; localities in area 2 (5–9, see Fig. 1 and Table 1) are from the outer folded belt exterior to the Variscan front and below the thrusted zone. The sites in area 3 (10 and 11) are within the Variscan belt and are sited above the two major thrust belts; they include repeat sampling at, or near to, locations investigated by earlier workers (McClelland Brown, 1983; Stearns and Van der Voo, 1987). Each locality comprises a number of distributed sites amounting to 53 in total; details are summarised in Table 1.

Samples were drilled in the field using a portable drill (with the exception of localities 2 and 9 where block sampling was initially undertaken and blocks were cored in the laboratory) and oriented by Sun and magnetic compasses. They were subsequently cut into 2.4 cm cylinders for routine palaeomagnetic processing.

Cores from green beds at sites 37 and 39 had very low intensities of magnetisation and were excluded from further study. A total of 453 cores from the remaining red sediments were subjected to progressive thermal demagnetisation in steps of 50° to 500°C and then in increments of 20° or 25°C to 700° or 720°C. "Minispin" magnetometers and Magnetic Measurements demagnetisers were employed, and the entire laboratory processing was conducted within a continuously monitored field-free space provided by a set of large Rubens coils (Rubens, 1945). The component structure of the magnetic remanence was evaluated from the study of orthogonal vector plots; components were isolated interactively and calculated by principal component analysis (Sherwood, 1989).

3. Palaeomagnetic results

During progressive thermal treatment the susceptibilities of 45% of the samples (203 cores) were monitored after each step. No significant changes were observed below 500°C. At higher temperatures the coarser sandy lithologies, including those exhibiting a higher blocking temperature component, show little further change to 700°C. Samples from fine sand and silt lithologies however, tend to undergo increases in susceptibility by up to an order of magnitude; this is often accompanied by erratic directional and remanent intensity behaviour and defines the effective limit of thermal treatment. Since susceptibility is always increased by heating in air, it is apparent that new magnetite is being formed by the breakdown of argillaceous minerals in these lithologies (Sect. 5).

Minor low-temperature components are commonly present in area 3 (74% of the collection) but are less apparent to the northeast (43% in area 2 and 68% in area 1). They are removed by 300°C and often by 150°C; directions are therefore not always accurately defined by demagnetisation steps utilised here, and since most are evidently of viscous origin, they are not discussed further. The same characteristic remanence component (ChRM) dominates the spectrum at sites from areas 2 and 3 and has southerly declination. Behaviours are illustrated in Figs. 3–5; examples of single convergent components with this direction are shown from area 2 in Fig. 4 but are well represented elsewhere (e.g., 4-5 and 6-5 in Fig. 3). The orthogonal projections illustrate the essentially convergent nature of this component residing in hematite with small perturbations in direction over limited temperature intervals; this effect, and the distributed blocking temperature spectra, are likely to be the signature of stages of hematite growth within the interval of diagenesis recorded by this mineral.

Principal components calculated from this segment of the spectrum are plotted area by area in Fig. 6 and group means calculated at sample level are listed in Table 2. This magnetisation is of uniform reversed polarity and is essentially a post-folding phenomenon in areas 1 and 2 where tilt adjustment yields a negative test significant at the 95% confidence level (McElhinny, 1964). In both cases the inclination is very shallow negative and SSW-directed. In area 3 however, both in-situ and tilt-adjusted directions are different from group 1 and 2 sites and the simple fold test is not diagnostic (Table 2). Hence, neither the in-situ nor the fully tilt-corrected mean is applicable to this magnetisation within the orogen; this point is analysed in Sect. 4 below.

Mean remanence directions are rotated clockwise by ca. 11° between areas 1 and 2. This is also shown when data are meaned at locality level and the essential conclusion is not affected by the tilt-adjusted assumption (Table 3). It becomes prominent at localities 7–9 within 15 km of the Variscan front (Fig. 7). Hence this component, although in part of post-folding origin, was progressively rotated clockwise by later deformation across the Variscan belt. Clockwise rotation of the ChRM continues into area 3 where it con-

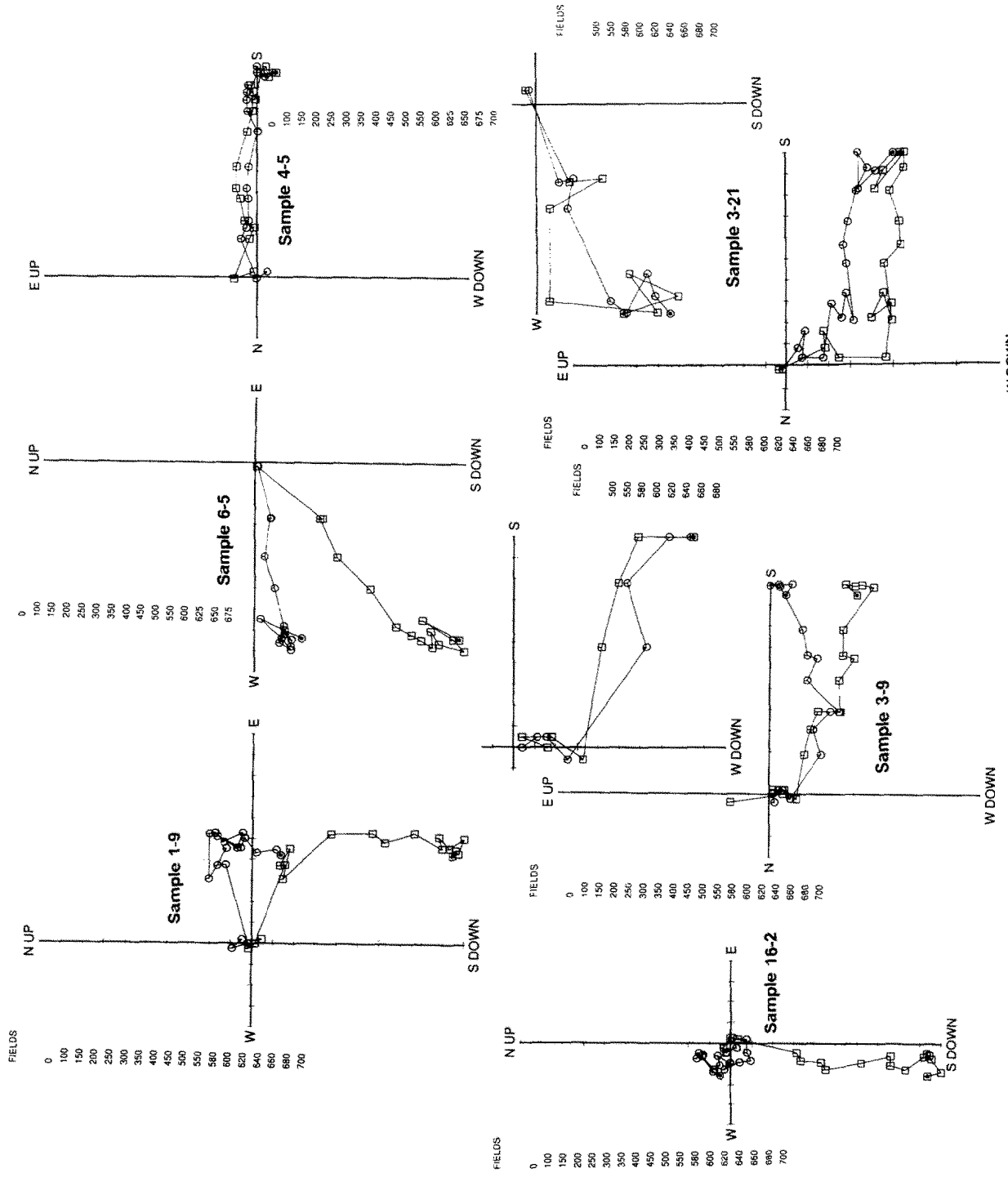


Fig. 3. Behaviour of sample magnetisations from area 1 (north of the Variscan orogenic front) to progressive thermal demagnetisation. The orthogonal projections show successive positions of the projection onto the horizontal plane (squares) and vertical plane (circles) with treatment steps (in °C) listed. Samples are referenced in terms of site and core numbers (Table 1). Intensities of magnetisation are mostly between 0.1 and 1×10^{-5} A m^2 kg^{-1}.

firms the essential conclusion of McClelland Brown (1983).

In area 3 this ChRM is convergent in the majority of samples (e.g., 50-5 and 53-7 in Fig. 5). In some cases it is not convergent and there is evidence for a residual high blocking temperature component (e.g., 49-1 and 52-1 in Fig. 5). In all these examples, behaviour became erratic at temperatures in the range 600–640°C (e.g., sample 52-1) and it proved impossible to reliably resolve the direction of this higher blocking temperature (hbt) component. It is, therefore, considered that there is no reliable evidence for preservation of a residual hbt component in this area (see also Stearns and Van der Voo, 1977).

In area 2, and more extensively in area 1, however, coherent hbt components are recovered following removal of the south-southwest shallow magnetisation in 7 samples from area 2 (3%) and 45 samples (30%) from area 1. Components in area 2 show no apparent groupings before or after tilt adjustment and the sample is too small for meaningful interpretation, but in area 1 southwest positive-directed hbt components are similar to the hbt components recognised more prevalently close to the Acadian front by Channell et al. (1992). Examples in Fig. 3 show both normal (e.g., 1-9) and reversed (3-9 and 3-21) polarities mostly recovered between 620° and 700°C. They are, therefore, likely to reside in grains of specular hematite although an origin in authigenic grains cannot be excluded (see for example, Turner, 1979). A composite source is suggested by the presence of a few lbt components with this direction at site 16 (e.g., 16-2 in Fig. 3). Generally, hbt components are sporadically preserved in one or two samples from a site, but are better represented at sites 3, 15 and 16 (Table 2). Overall grouping is improved (but not significantly so in this case) following tilt adjustment and five examples then have normal polarity (Fig. 8). The mean reversed axis (D/I = 246/38°, Table 4) compares with the pre-Acadian hbt component resolved by Channell et al. (1992,

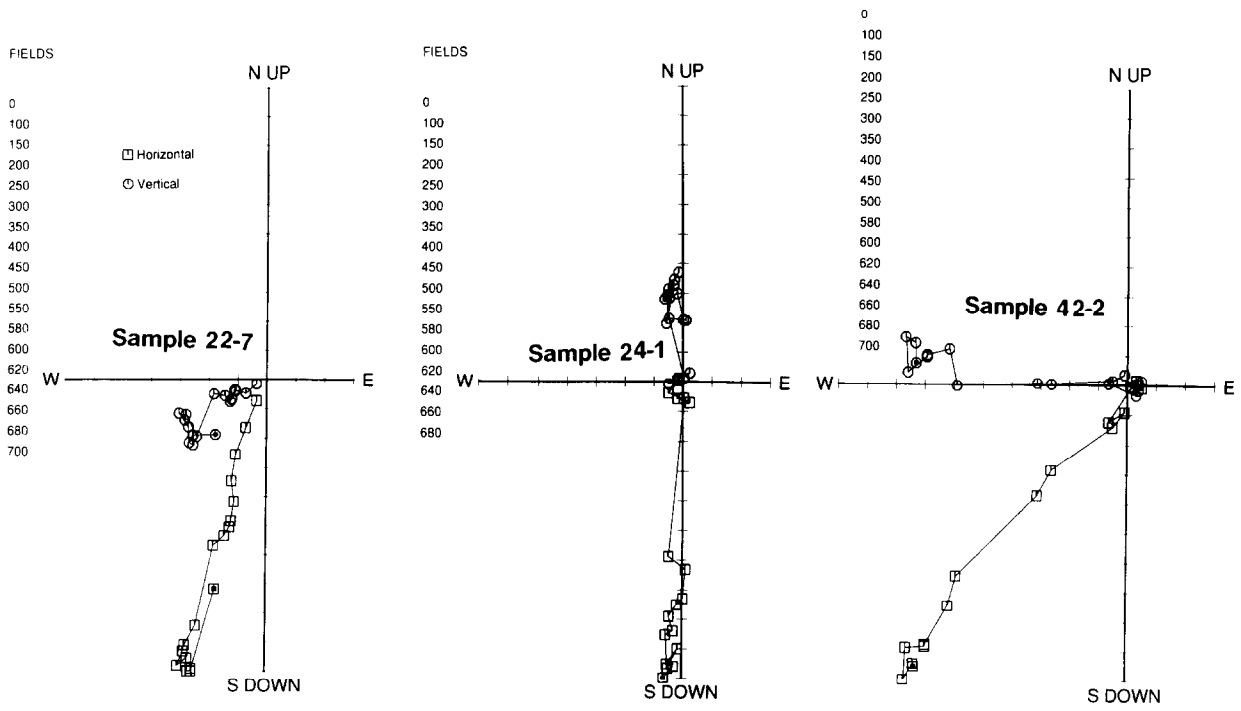


Fig. 4. Behaviour of sample magnetisations from area 2 (bordering the Variscan orogenic front) with progressive thermal demagnetisation. Symbols are as for Fig. 3.

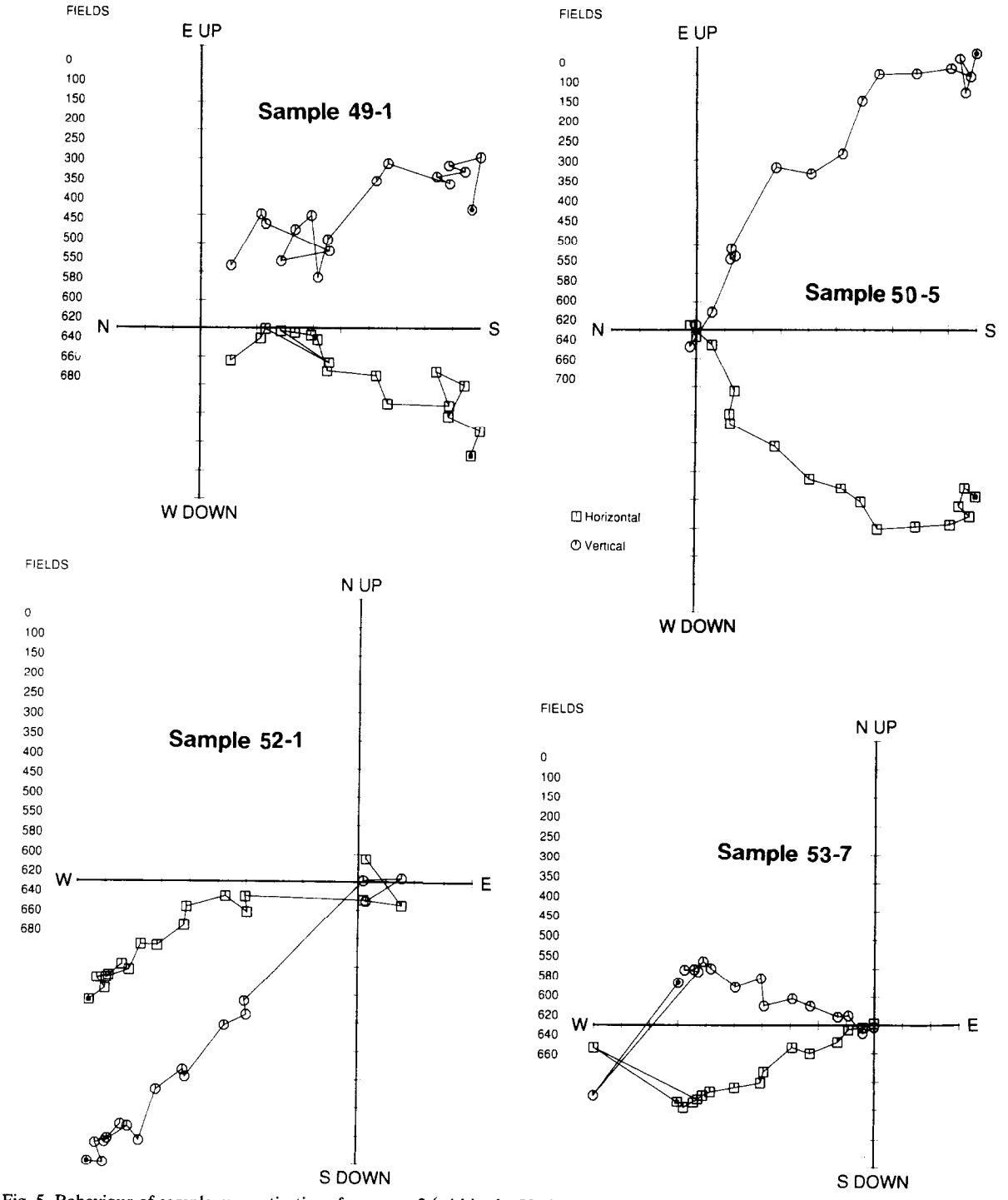


Fig. 5. Behaviour of sample magnetisations from area 3 (within the Variscan orogenic front) with progressive thermal demagnetisation. The symbols are as for Fig. 3.

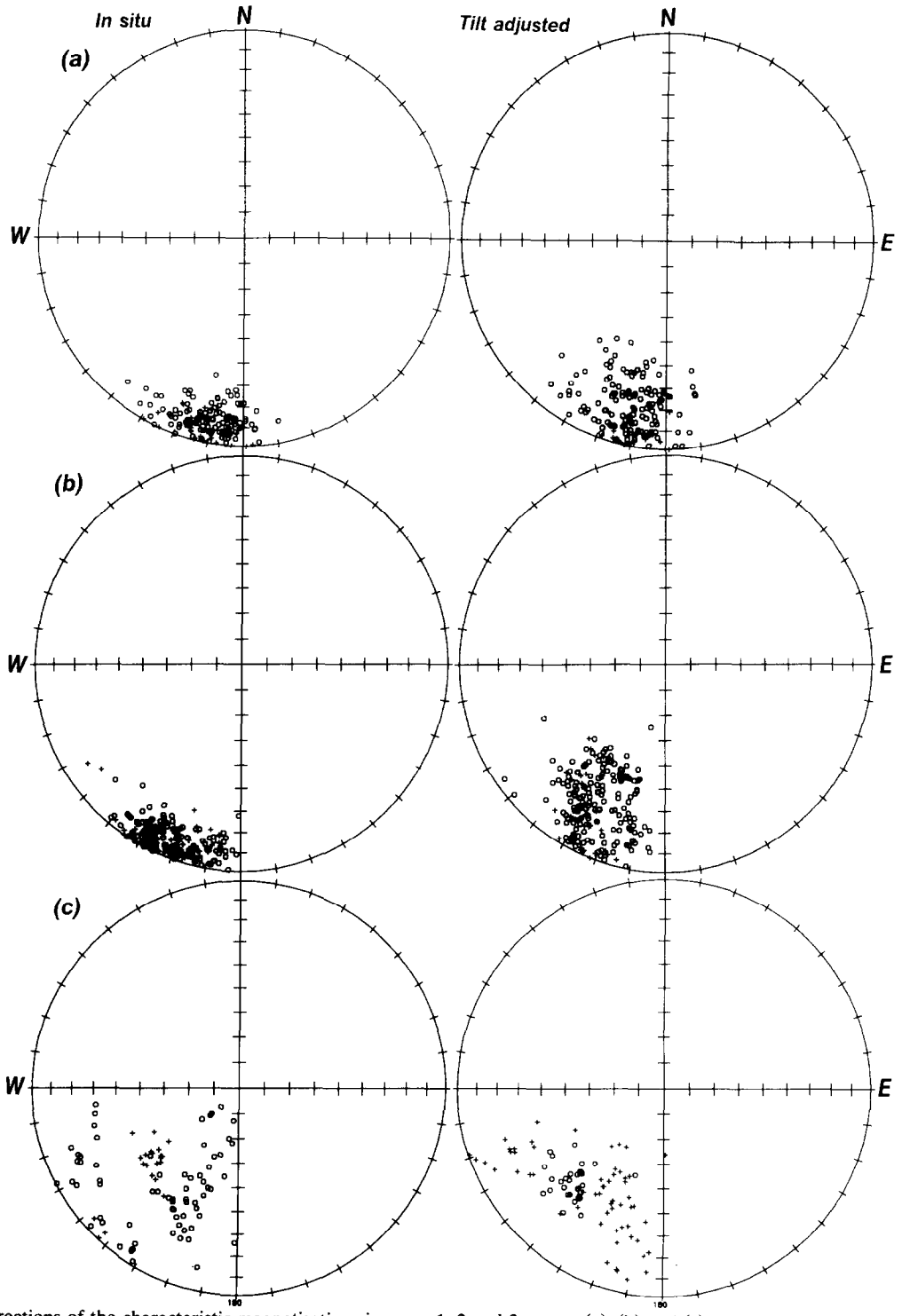


Fig. 6. Directions of the characteristic magnetisations in areas 1, 2 and 3—resp. (a), (b) and (c)—before and after tilt adjustment. Crosses are lower-hemisphere projections and circles are upper-hemisphere projections; equal area plots. The statistics for these distributions are given in Table 2.

Table 2
Palaeomagnetic results for medium blocking temperature components

Site No.	N	R	k	α_{95}	D (in situ)	I	D (tilt adjusted)	I
Area 1, north of the Variscan belt								
1	12	11.69	35.2	7.4	186.5	−2.7	196.5	−28.6
2 (i)	8	7.72	25.1	11.3	193.2	−4.3	199.4	−22.7
2 (ii)	4	3.86	21.3	20.4	141.3	−30.8	157.6	−62.2
3 (i)	20	19.72	68.2	4.0	190.0	−5.3	190.0	−5.3
3 (ii)	10	9.44	16.4	12.4	237.7	36.5	238.1	35.8
4	5	4.96	111.8	7.3	197.4	−6.7	179.0	−26.7
5	7	6.86	41.5	9.5	194.8	−5.8	195.5	−22.3
6	6	5.85	32.8	11.9	203.6	−10.3	207.4	−16.2
7	5	4.86	28.5	14.6	194.5	−2.1	198.0	−19.2
8	5	4.96	101.9	7.6	192.7	−12.7	195.3	−25.9
9	7	6.90	59.9	7.9	188.7	−6.8	190.6	−23.5
10	5	4.98	162.1	6.0	184.6	−14.1	186.6	−28.1
11	5	4.98	214.2	5.2	184.4	−10.4	186.0	−24.5
12	5	4.82	22.8	16.4	190.4	−16.2	193.3	−29.5
13	5	4.95	79.5	8.6	188.2	−16.1	190.8	−29.8
14	6	5.95	99.1	6.8	187.1	−19.9	194.1	−34.4
15 (i)	15	14.67	42.4	5.9	189.8	−4.2	189.2	−4.9
15 (ii)	3	2.97	66.3	15.3	245.8	28.7	243.7	30.6
16 (i)	7	6.79	28.7	11.5	189.1	−3.0	189.3	3.4
16 (ii)	5	4.79	19.2	17.9	259.9	48.2	257.5	46.4
17	9	8.83	48.0	7.5	201.9	−8.6	205.7	−11.3
18 (i)	11	10.78	44.6	6.9	190.4	−10.0	191.5	−12.9
18 (ii)	7	6.75	24.4	12.5	220.4	−56.7	237.2	−55.2
19 (i)	8	7.96	179.6	4.1	186.7	−12.1	187.3	−15.7
19 (ii)	5	4.86	28.3	14.6	201.8	−47.5	208.3	−49.0
Area 2, northern margin of the Variscan belt								
20	7	6.95	124.4	5.4	195.9	7.3	206.5	−40.8
21	10	9.90	92.7	5.0	194.1	16.8	200.8	−35.7
22	10	9.92	114.7	4.5	196.2	10.2	207.8	−39.5
23	11	10.89	95.3	4.7	186.4	−6.5	189.7	−17.2
24	9	8.94	126.7	4.6	188.5	−8.1	193.6	−17.2
25	7	6.98	393.6	3.1	192.1	−15.5	196.2	−23.6
26	5	4.97	125.5	6.9	201.5	−7.4	205.3	−44.7
27	9	8.95	160.0	4.1	197.3	−3.7	199.3	−41.5
28	14	13.82	72.3	4.7	199.8	−13.4	213.2	−44.7
29	18	17.56	38.3	5.7	214.0	2.7	219.8	−25.5
30	6	5.97	152.9	5.4	206.6	5.4	211.1	−26.3
31	5	4.99	854.6	2.6	210.4	−1.4	212.2	−14.4
32	5	4.99	374.6	4.0	208.8	1.2	209.0	−8.3
33	5	4.97	131.3	6.7	209.4	3.6	218.3	−30.9
34	5	4.99	421.3	3.7	206.8	−12.8	207.4	15.1
35	3	2.99	553.0	5.3	203.1	−5.0	217.9	32.4
36	5	4.99	586.5	3.2	208.6	−3.4	216.3	30.2
38	5	4.99	358.3	4.1	197.7	−2.9	218.4	45.5
40	8	7.97	223.1	3.7	206.9	−13.5	208.4	17.0
41	18	17.94	261.9	2.1	208.0	−6.3	212.9	−24.3
42	2	1.99	121.4	–	215.5	−4.8	217.1	−16.1
43	7	6.88	48.3	8.8	204.2	−10.5	204.2	−10.5
44	8	7.98	315.5	3.1	202.7	−3.0	202.7	−5.6
45	7	6.95	121.5	5.5	203.5	−2.9	203.5	−2.9
46	12	11.75	43.4	6.7	202.9	−9.7	202.9	−9.7
48	9	8.92	101.1	5.2	210.0	−12.8	207.0	1.3

Table 2 (continued)

Site No.	N	R	k	α_{95}	D (in situ)	I	D (tilt adjusted)	I
Area 3, within Variscan belt								
49	24	22.68	17.4	7.3	200.3	−52.2	196.3	36.0
50	19	18.73	67.2	4.1	208.8	−38.6	208.9	38.8
51	13	12.79	57.4	5.5	216.6	−6.3	228.3	−39.9
52	18	17.87	129.1	3.1	230.2	45.2	221.8	−34.0
53	16	15.58	35.3	6.2	244.8	−17.2	246.3	21.7
Group mean results								
Group 1, 151 samples								
		147.2	39.2	1.9	191.1	−8.0		
(tilt adj.:		144.6	23.4	2.4			192.5	−16.7)
Palaeomagnetic pole: 341.1°E, 41.2°S, $d_p/d_m = 0.9/1.9°$								
Group 2, 210 samples								
		204.2	36.2	1.6	202.3	−3.8		
(tilt adj.:		192.4	11.9	3.0			207.0	−18.2)
Palaeomagnetic pole: 327.6°E, 36.6°S, $d_p/d_m = 0.8/1.6°$								
Group 3, 90 samples								
In situ:		69.8	4.4	8.1	218.4	10.5		
Tilt adj.:		69.9	4.4	8.0			220.1	−19.4

D and I are the mean declination and inclination, respectively, derived from a site population of N samples yielding a resultant vector of magnitude R and a cone of 95% confidence about the mean direction of radius α_{95}; k is the Fisher precision parameter ($= (N-1)/(N-R)$); d_p and d_m are the radii of the oval of 95% confidence about the pole position in the colatitude direction and at right angles to it, respectively.

$D/I = 232/32°$). Sample locations in this study are from either side of the Neath disturbance where deformation is post-Devonian in age and may be entirely Variscan or post-Variscan in age (R. Gayer, pers. commun., 1990). Hence it is not possible to resolve the age of this remanence, although by analogy with the study of Acadian structures to the west, it is also likely to be Lower Devonian and pre-Acadian in origin. The implication to the apparent polar wander path is that a westerly excursion of the pole occurred in Late Silurian and Early Devonian times; this is now

Table 3
Locality mean magnetisations

Locality	Samples	R	k	α_{95}	D (in situ)	I	D (tilt adjusted)	I
1	40	39.09	42.8	3.5	189.6	−4.3	193.6	−15.7
2	61	59.90	40.0	2.9	191.6	−10.9	192.9	−25.5
3	22	21.46	38.9	5.0	189.6	−3.9	189.2	−2.3
4	28	27.40	45.1	4.1	193.0	−10.2	194.8	−13.3
5	27	26.71	90.5	2.9	195.4	11.8	204.4	−38.6
6	27	26.74	99.8	2.8	188.6	−9.4	192.6	−18.9
7	28	27.65	76.9	3.1	199.3	−9.2	207.3	−44.2
8	24	23.48	44.4	4.5	212.1	3.4	217.6	−25.8
9	102	100.75	80.8	1.5	206.2	−6.5	206.2	−6.5
10	56	51.38	11.9	5.8	208.3	−37.1	207.8	22.0
11	34	28.43	5.9	11.1	238.1	16.5	223.9	−8.4

The statistics are for in-situ groupings.

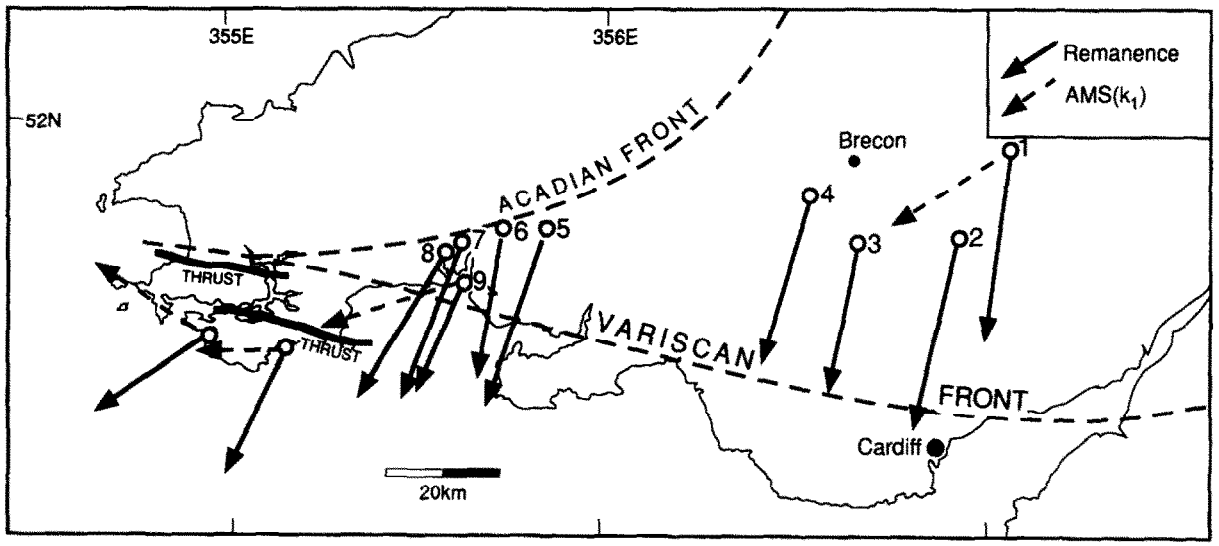


Fig. 7. In-situ locality mean directions illustrating the clockwise rotation of the characteristic magnetisation vector towards the west in South Wales; although the magnetisations appear to be predominantly syn-folding (see text), tilt adjustment does not appreciably affect this general picture of clockwise rotation and 95% error limits are mostly less than 5° (see Table 3). Also shown are in-situ directions of maximum susceptibility (k_1) at sites 1, 9, 10 and 11 (see Fig. 2).

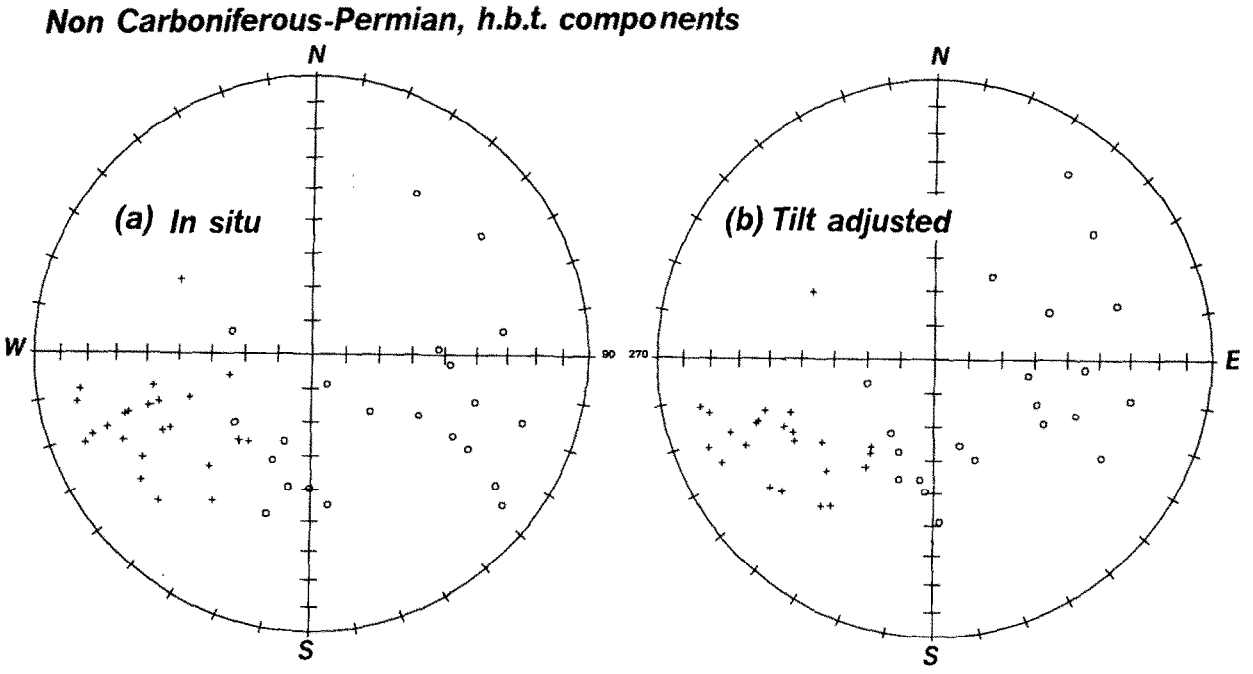


Fig. 8. Medium-high blocking temperature components of magnetisation recovered from samples in area 1 which do not belong to the populations of Upper Carboniferous–Permian components plotted in Fig. 6. Symbols are as for Fig. 6.

Table 4
Medium-high blocking temperature components of non-Carboniferous–Permian age at Group 1 sites

N	R	k	α_{95}	D (in situ)	I	D (tilt adjusted)	I
(i) 27	25.19	14.4	7.6	245.7	38.7		
(i) 27	25.34	15.7	7.2			246.3	38.4
(palaeomagnetic pole: 297.3°E, 3.3°N, $d_p/d_m = 5.1/8.6°$)							
(ii) 17	15.53	10.9	11.2	202.5	−57.4		
(palaeomagnetic pole: 298.2°E, 69.1°S, $d_p/d_m = 12.1/19.1°$)							
(ii) 17	15.27	9.2	12.4			177.3	−57.3

recorded both across Britain north of the Variscan front (Channell et al., 1992), and in continental Europe (Piper and Poppleton, 1991).

The remaining high blocking temperature components in area 1 have negative inclinations with a range of southerly declinations (Fig. 8). With one exception, they are recorded only at three sites (2, 18 and 19) for which means are listed in Table 2. They are not obviously a single population and tilt adjustment is not diagnostic (Table 4). However, the in-situ mean ($D/I = 203/-57°$) is compatible with a later Mesozoic reversed field in this area (Fig. 9); this suggests authigenic hematite growth linked to mineralising brine migrations and associated with sedimentary basin development in adjoining regions during these times.

4. The relationship between folding and regional magnetic overprinting

Simple tilt adjustment of the characteristic remanence is unable to resolve the origin of this remanence within the Variscan belt (area-3 sites).

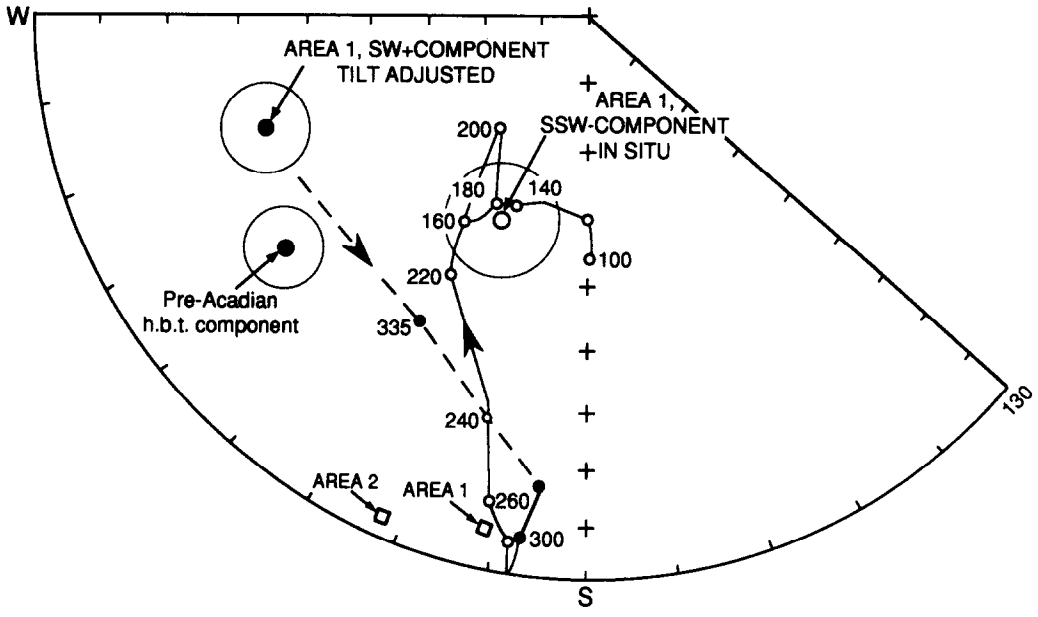


Fig. 9. The path of the reversed palaeofield direction in South Wales between 310 and 100 Ma predicted from the Eurasian apparent polar wander path (Piper, 1987). The ChRM mean directions of the sites in areas 1 and 2 (see group 1 and 2 calculations in Table 2) are shown. Also shown with 95% confidence circles is the mean direction of the residual hbt components identified in area 1 sites (Table 4) and the mean pre-Acadian remanence resolved from Old Red Sandstones near the Acadian front (see Fig. 2) by Channell et al. (1992). The pre-310 Ma palaeofield path for Britain is not yet well defined but the predicted direction for ca. 335 Ma during the Early Carboniferous is shown for reference after Piper et al. (1992).

Furthermore, neither mean direction (Table 2) is compatible with the palaeofield in this area: it conforms with the post-Variscan path but only in the tilt-adjusted configuration, and with the pre-Variscan path but only in the in-situ position (cf. Table 2 and Fig. 9). It therefore seems likely that: (1) acquisition occurred at some intermediate stage of folding (Chamalaun, 1964; McClelland Brown, 1983); and/or (2) that it is influenced by plunge corrections.

The program FOLDCONT (Setiabudidaya et al., in prep.) has been developed to evaluate this kind of problem. It incrementally untilts the limbs of a fold (in two-degree steps) and introduces the option of untilting each limb separately. The precision parameter, k, is compared with the value in the in-situ position; values of k over the whole range of possible tilt adjustments are contoured. This test can simultaneously be compared (e.g., Figs. 10 and 11) with the one-dimensional fold test in which both limbs of the fold are untilted together; the range of k at which the null hypothesis of McElhinny (1964) cannot be rejected at the 95% confidence level is conveniently illustrated on the same diagrams (e.g., Miller and Kent, 1986).

It is not always clear that fold geometries or palaeomagnetic directions, derived from the maximum k value achieved during progressive unfolding provide the correct solution. The direc-

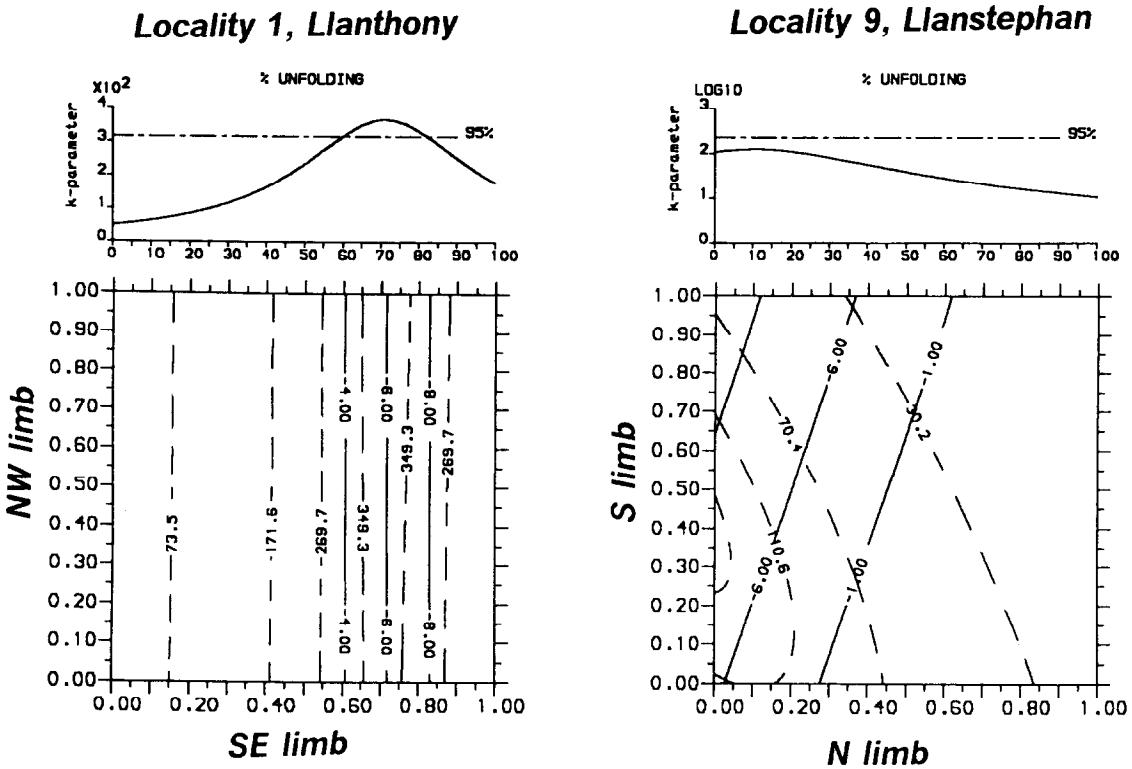


Fig. 10. Results of palaeomagnetic fold tests applied to structures at localities 1 (Llanthony) and 9 (Llanstephan). In each case the upper figure illustrates the value of the precision parameter, k (logarithmic scale), as the structures are symmetrically untilted. The minimum k value satisfying the null hypothesis at the 95% confidence level (McElhinny, 1964) is shown as the dashed line. The lower figures illustrate results of the FOLDCONT program which evaluates all possible geometries of untilting in 2° steps and contours the resultant field of precision, k (dashed lines). The values of untilting which yield the inferred ancient palaeofield inclination during the remagnetisation event are shown by the solid lines.

tion may, for example, be incompatible with geological constraints (e.g., Hudson et al., 1989). Accordingly, FOLDCONT has the option of introducing a predicted palaeofield direction and associated error bandwidth into the analysis. This option is useful in the present instance because it permits geometry of the fold structures to be recovered at the time of magnetisation in a well-constrained ancient field direction. The expected inclination with upper and lower error limits appears as straight lines superimposed onto the contour plots.

The Variscan overprint overwhelmingly recorded in Old Red Sandstones has a south-southwest declination with shallow negative inclination. The same coherent overprint is also ubiquitous in overlying Westphalian 'A' Coal Measures ($D/I = 195.4/-5.1°$, Setiabudidaya et al., 1994) and is compatible with the regional palaeofield direction predicted from the Eurasian APW path at ca. 280 Ma (Fig. 9). Accordingly, a mean inclination of $-6 \pm 5°$ is used in FOLDCONT to recover fold geometries imparted during deformation within the orogenic belt during these times. Results from locality 1 (Llanthony) across the Neath disturbance where the plunge is negligible (1/228°E) are illustrated in Fig. 10. The null hypothesis of McElhinny (1964) cannot be rejected between 60 and 83% unfolding (Fig. 10, left) but since the northwestern limb has a dip of only 2°, the contoured k values in Fig. 10 are only sensitive to unfolding of the southeastern limb; the maximum k value in this case corresponds to the predicted field inclination of $-6°$ at 76% unfolding (Table 5).

The fold at locality 9 (Llanstephan) is an open anticline with small plunge (5/250°E). The northern limb has to be left in situ and the southern limb unfolded by 36% to optimise the k parameter at which point the mean direction ($D/I =$

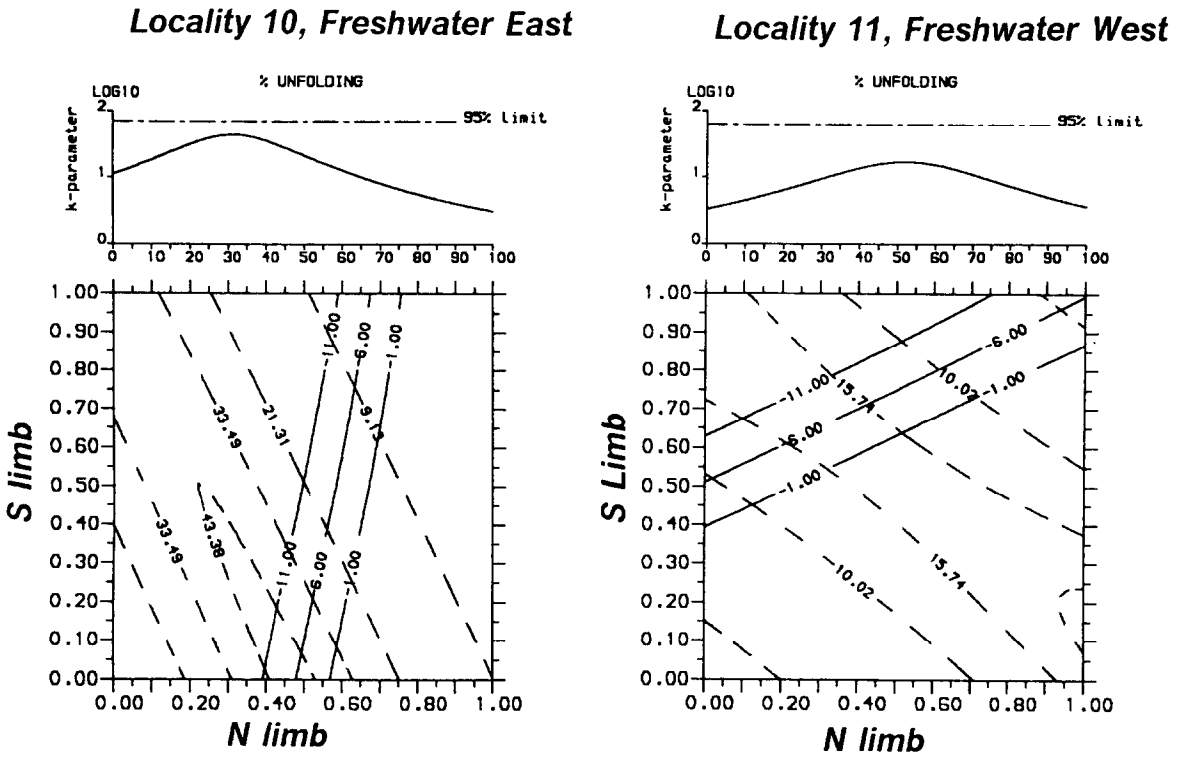


Fig. 11. The results of palaeomagnetic fold tests applied to structures at locality 10 (Freshwater East) and locality 11 (Freshwater West, Castlemartin Corse). The symbols etc., are as for Fig. 10.

Table 5
Results of palaeomagnetic fold tests

Locality	Stepwise unfolding of each limb					Symmetrical unfolding: in situ				Symmetrical unfolding: plunge adjusted			
	No. of sites	Dip of N. limb	Dip of S. limb	D/I	k	% unfolded	D/I	α_{95}	k	% unfolded	D/I	α_{95}	k
Llanthony (1)	3	0	11	188.6/−6.1	368 [a]	72	188.6/−6.1	2.2	366 [a]	10	206.3/−8.6	1.9	121
Llanstephan (9)	11	49	16	207.3/−9.1	157	10	206.8/−5.2	1.9	124	36	201.9/−14.9	7.9	30
Freshwater E. (10)	3	43	41	211.5/−1.1	113 [a]	32	205.9/−18.8	6.5	44	34	212.1/−17.4	4.1	107 [a]
Freshwater W. (11)	2	38	31	233.8/−10.8	71 [a]	52	228.6/4.1	14.1	17	54	233.0/1.1	7.1	65 [a]
Anticline (10+11)	3	43	44	214.4/−1.4	49 [a]	56	204.1/1.1	5.8	54 [a]	54	214.4/1.8	6.1	49 [a]
Syncline (10+11)	2	38	31	229.1/−10.8	25 [a]	20	230.4/−11.7	12.9	20	30	229.1/−10.8	11.6	25

[a] Passes the null hypothesis at the 95% confidence level (McElhinny, 1964).

207.3/−9.1°) is within the reference bandwidth and also close to the mean remanence from the hinge area of the fold ($D/I = 202.3/-5.6°$). Simultaneous unfolding (Fig. 10) optimises k at 10% unfolding but yields a negative test implying that the remanence here was acquired after most, or all, of the folding was completed. Furthermore, plunge correction reduces the k values and suggests that a plunge was already present before the magnetising event occurred. Comparison of the fold test results from localities 1 and 9 (Table 5), therefore, implies that the magnetising event largely postdated folding close to the orogenic front but largely predated it on the platform remote from the front. This is palaeomagnetic support for regional geological evidence identifying a northward migration of the climax to the Variscan Orogeny (Owen, 1974).

The folds within the Variscan belt at localities 10 and 11 are more complex structures. The plunges of the Freshwater East and Freshwater West (or Castlemartin Corse) anticlines are calculated from cleavage–bedding intersections (Freshwater East, northern limb 8/297°E, southern limb 14/95°E; Freshwater West, northern limb 7/275°E, southern limb 15/310°E; P.L. Hancock, pers. commun., 1991). Four plunges are used because limbs are shared between the anticlines and a complementary plunging syncline (the Orielton syncline, Fig. 2); thus analysis considers the structures individually and then as an integral whole (Table 5).

At Freshwater East the northern limb must be left in situ and southern limb unfolded by 46% to achieve optimum grouping; the remanence then also lies within the reference bandwidth ($D/I = 204.8/-7.1°$, $\alpha_{95} = 6.4°$) showing that fold shape was more symmetric when the characteristic remanence was acquired. Adjustment for plunge enhances the grouping and yields a vector more rotated in a clockwise sense ($D/I = 211.5/-1.1°$, Table 5). A similar solution is derived at Freshwater West: the northern limb is unfolded less (22%) than the southern limb (72%) for optimum grouping; the mean ($D/I = 229.3/-10.9°$, $\alpha_{95} = 13.9°$) is within the reference bandwidth but grouping is enhanced and the vector rotated clockwise (Table 5) when adjustment for plunge is applied. Treatment of the combined structure yields substantially the same conclusions (Table 5).

The results of Table 5 demonstrate the need to separately untilt the limbs of folded structures when investigating the relationship between folding and magnetisation: in these examples the expected palaeofield directions are only achieved after untilting the southern limb more than the northern limb. Hence, symmetrical fold structures, already imparted to these rocks at the time of the magnetising event, have been steepened up from the south during succeeding stages of Variscan tectonism during which they also acquired a plunge. The general clockwise rotation of the remanence vector to the west in South Wales noted by McClelland Brown (1983, and see also Fig. 7) is confirmed but applies only to a total Variscan overprint. The largest rotation is concentrated within the zone between Freshwater East and Freshwater West (Fig. 2 and Table 5); this is evaluated in Sect. 6.

5. Isothermal remanent magnetisations and magnetic fabrics

Old Red Sandstone sediments are only rarely saturated by 3000 mT and isothermal remanent magnetisations (IRM's) are therefore dominated by hematite (Fig. 12). There is little apparent difference in the coercivities of samples from the three areas, but samples from areas 2 and 3, which are dominated by folded Variscan overprints, have a more prominent bimodal character with a lower coercivity phase saturating by 2000 mT. If the causative mineral is magnetite it is present in small amounts because only rarely (e.g., sample 47 in Fig. 12) does it dominate the spectrum although the saturation remanence of hematite is only about 0.5% of that of magnetite. Most samples become magnetically harder following thermal demagnetisation (Fig. 12) even when there is little change in susceptibility.

The origin of the causative minerals can be examined by firstly applying a high field IRM (in this case 3000 mT) in the vertical direction, secondly applying a low field IRM (in this case 300

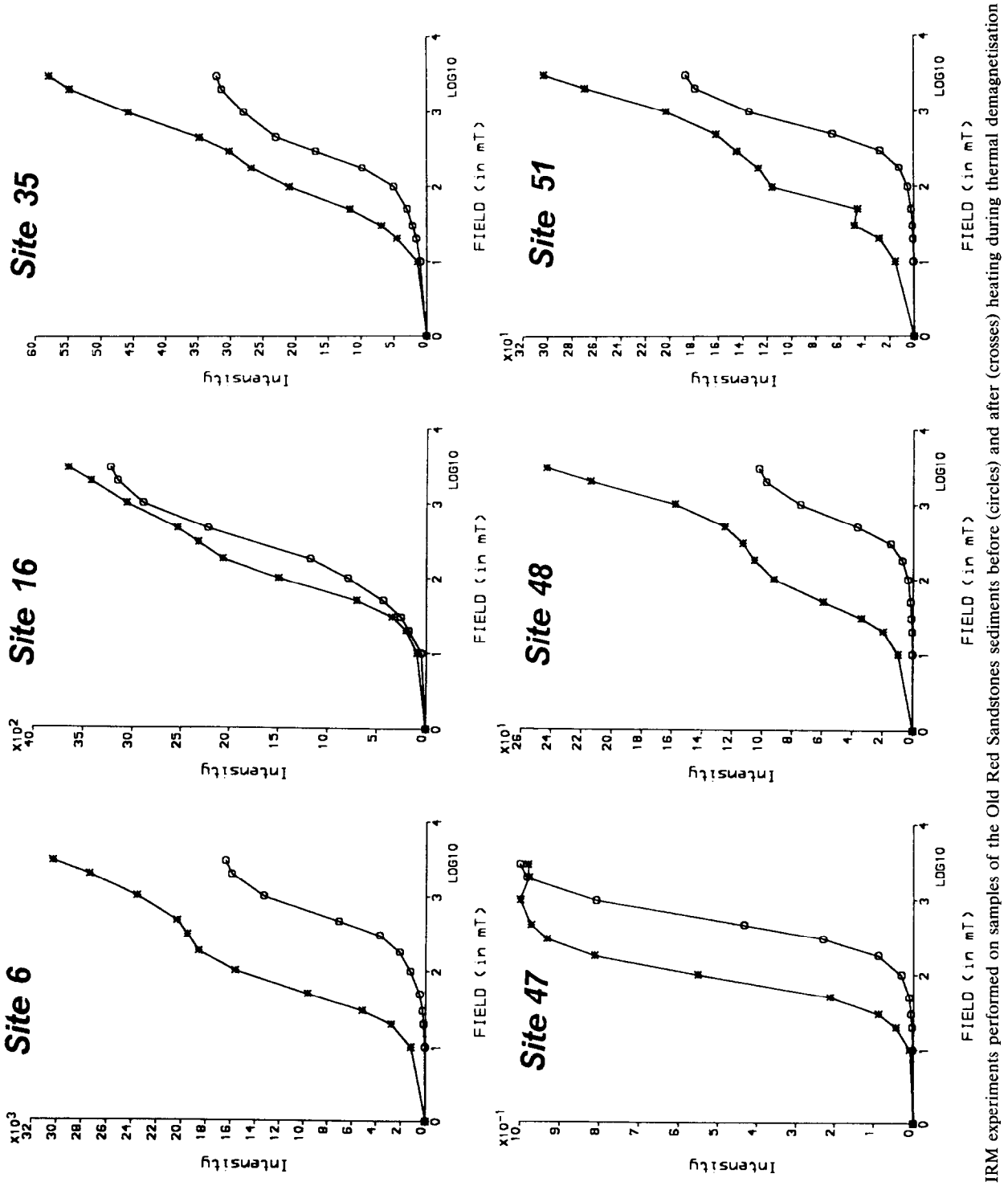


Fig. 12. IRM experiments performed on samples of the Old Red Sandstones sediments before (circles) and after (crosses) heating during thermal demagnetisation experiments.

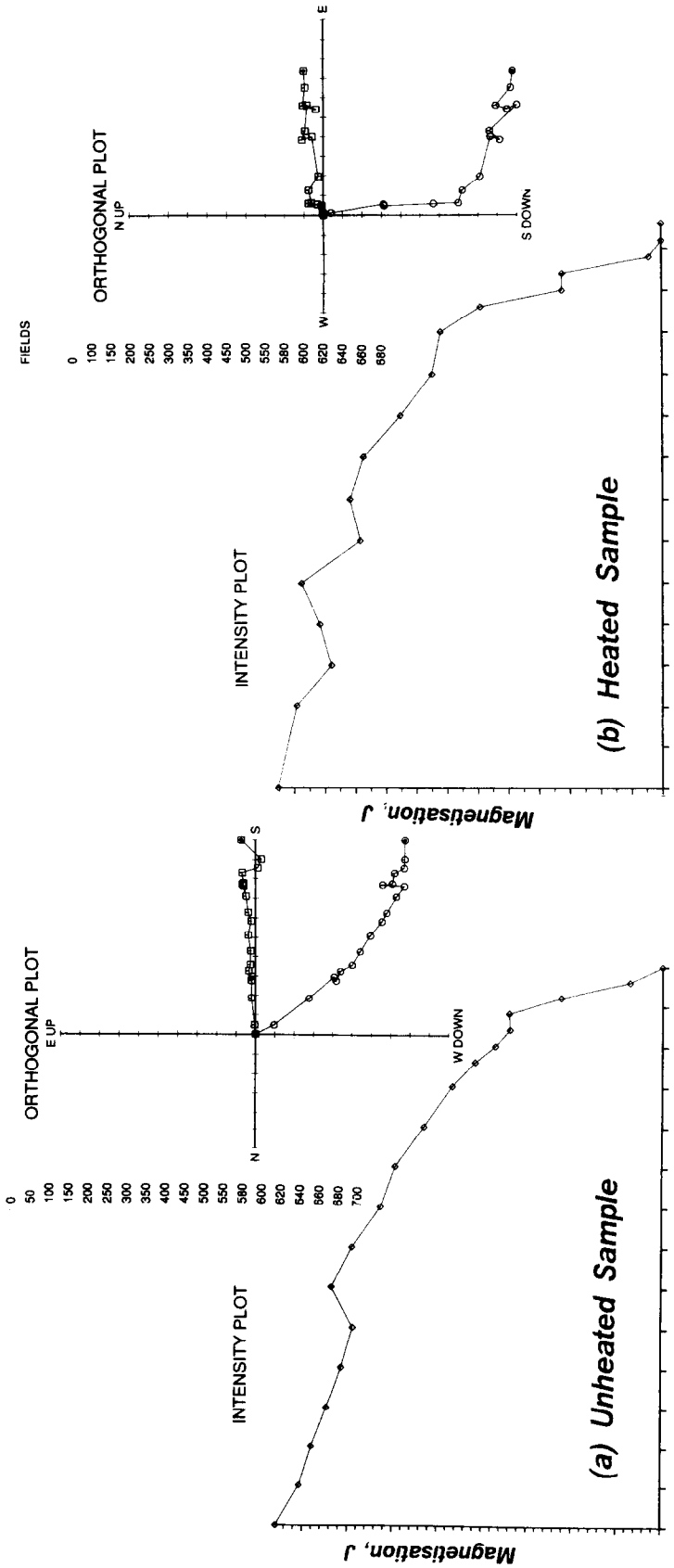


Fig. 13. Thermal demagnetisation results from (a) untreated and (b) thermally demagnetised samples from the Old Red Sandstones following application of a high field IRM (3000 mT in a vertical direction) and a low field IRM (300 mT in a horizontal E–W direction). Note that magnetite most receptive to saturation in the second applied field is more prominent in the heated sample. Symbols on the orthogonal projections are as for Figs. 3–5. The intensities of magnetisation of the samples illustrated are 86 and 125×10^{-6} A m^2 kg^{-1}, respectively.

mT) in a horizontal direction, and then thermally demagnetising the resultant remanence (Fig. 13). Most typically the resultant component has an intermediate direction. In unheated samples this is subtracted fairly uniformly over the blocking temperature spectrum to the Curie point of hematite which has partially saturated in both high and low fields; the broad coercivity spectra are mirrored in broad blocking temperature spectra (Figs. 12 and 13; Dunlop, 1972). Thermally treated samples, however, show contrasting behaviour: the first component is subtracted more or less discretely by the Curie point of magnetite, while the residual (dominated by the vertical-directed high-field IRM in these examples) is only subtracted by the Curie point of hematite.

Thus, the magnetic effect of thermally demagnetising these Old Red Sandstone sediments has been to produce small quantities of secondary magnetite. By conducting the demagnetisation in a nullified field environment, this alteration is unable to contribute to the remanence spectrum. Hence secondary magnetite is not the cause of multicomponent behaviour observed during the demagnetisation because: (1) with the exception of the low blocking temperature components of probable viscous origin, second components are observed in only a small fraction of the collection and are mostly within the higher hematite range (550–700°C); (2) most specimens showing this behaviour are from area 1 which proved to be most resistant to chemical change during laboratory treatment. It is possible that components reported by earlier workers (Chamalaun and Creer, 1964—these authors declined to calculate a pole position because of the large distance between their demagnetising and measuring facilities; McClelland-Brown, 1983) which this study has failed to replicate are IRM's acquired between demagnetisation and measurement.

Anisotropy of magnetic susceptibility (AMS) is a tensor which can be represented as a triaxial ellipsoid with maximum (k_1), intermediate (k_2) and minimum (k_3) principal axes. It was determined using a 'Minisep' anisotropy delineator for samples from localities 1, 9–11; these are representative of the three areas (Fig. 2) and permit possible differences between the strongly and

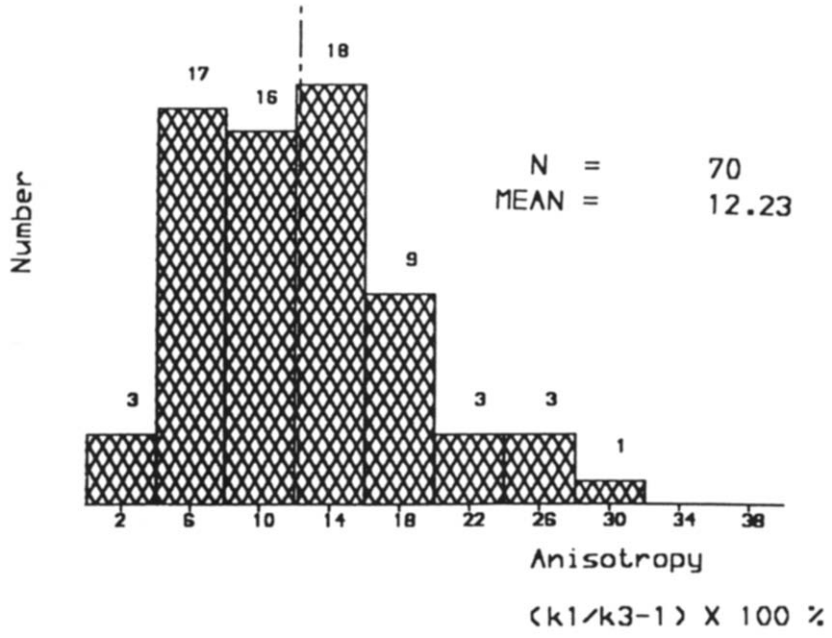


Fig. 14. Histogram of the magnitude of susceptibility $[(k_1/k_3)^{-1}]$ values at localities 1, 9, 10 and 11 in Old Red Sandstone sediments.

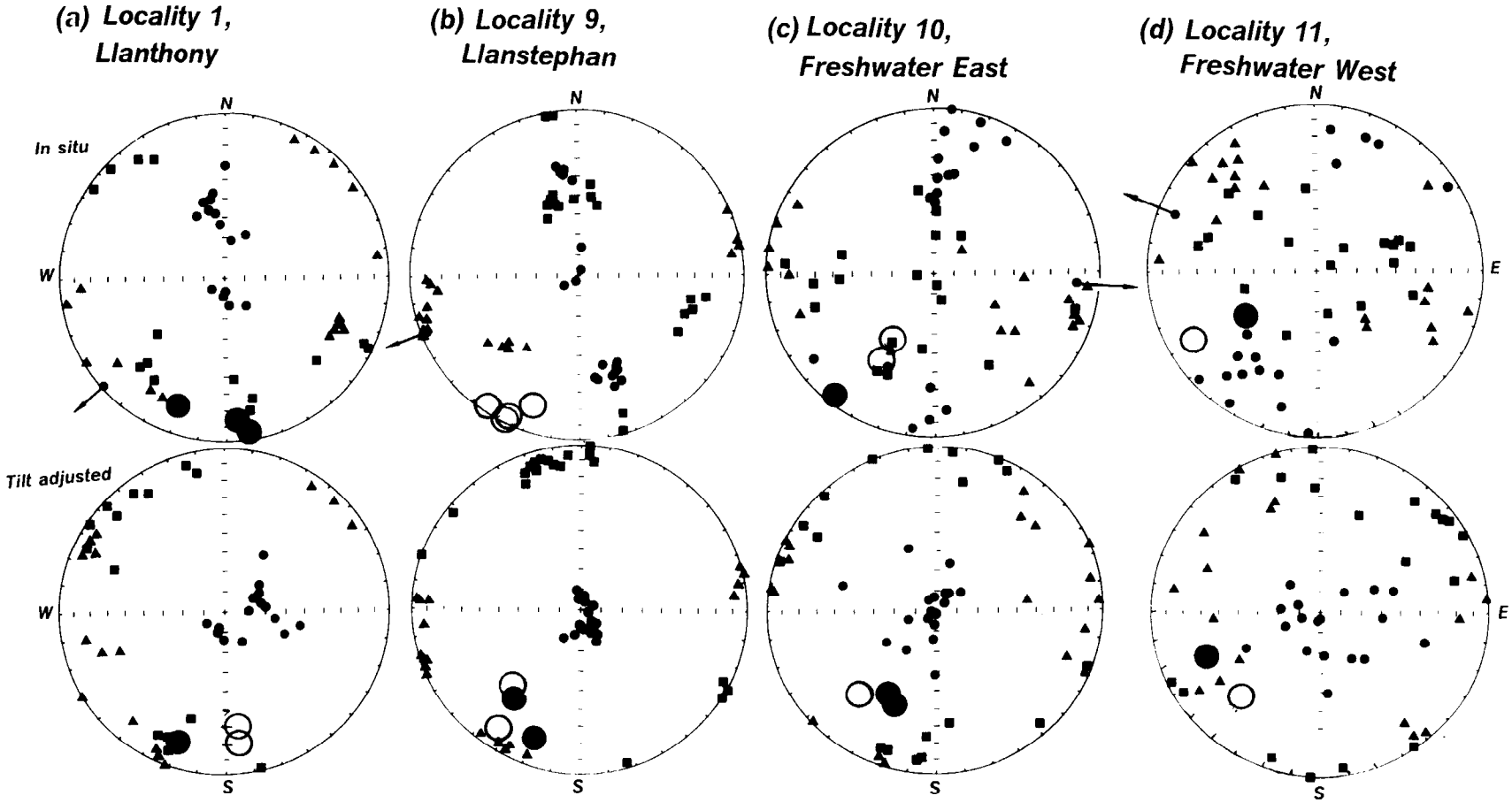


Fig. 15. The directions of principal susceptibility axes at localities 1, 9, 10 and 11. The upper projections are in situ and the lower projections are tilt adjusted. The symbols refer to the maximum (k_1, triangles), intermediate (k_2, squares) and minimum (k_3, dots) axes of the ellipsoids. The large symbols are mean remanence directions. The orientation of the fold axis with the direction of plunge is indicated by the arrow in the in-situ configuration; equal area projections.

weakly deformed sites to be monitored. The average magnitude of anisotropy (k_1/k_3) is 12% although a few samples have magnitudes ranging up to 32% (Fig. 14). Ellipsoids are predominantly oblate but there is an appreciably higher proportion of prolate ellipsoids at locality 11 where the lithologies are fine-grained argillaceous compared with the arenaceous lithologies tested elsewhere. The magnitude of the anisotropy tends to be about 6% higher in the beds strongly deformed by the Variscan episode (Table 6).

Orientations of the principal susceptibility axes are shown in Fig. 15. Large circles show mean directions of characteristic remanence at each site and the arrows (only shown on the projections in geographic coordinates) indicate strike of the fold and plunge of the axes. The fold axes of the Freshwater East (10) and Castlemartin Corse (11) anticlines are the average of four plunges determined on each limb and calculated from bedding–cleavage intersections (P.L. Hancock, pers. commun., 1991).

There is a tendency for the maximum susceptibility axes to be oriented parallel to fold axes but there is no relationship between susceptibility axes and mean remanence directions. Furthermore, susceptibility axes viewed in stratigraphic coordinates have minima clustering about the poles to the bedding (Fig. 15). Hence we conclude that the anisotropy remains dominated by primary bedding-related causes although this relationship becomes obscured at the more deformed sites as a result of Variscan strain. Thus it is least apparent at locality 11 where the maxima coincide closely with the fold axis and, in part, with the direction of plunge.

Table 6
Summary of AMS results

Locality	N	Magnitude (k_1/k_2)	Lineation (k_1/k_2)	Foliation (k_2/k_3)	% Oblate
1, Llanthony	15	1.073	1.020	1.053	93
9, Llanstephan	19	1.133	1.037	1.094	84
10, Freshwater E.	18	1.122	1.024	1.095	94
11, Freshwater W.	18	1.140	1.055	1.080	39

Table 7
Mean tilt-adjusted directions of maximum and intermediate axes at the four localities

Locality	k_1		k_2	
	D/I	α_{95}	D/I	α_{95}
1, Llanthony	245/8	19	155/−2	19
9, Llanstephan	246/3	11	156/−7	11
10, Freshwater E.	257/−8	16	167/−3	17
11, Freshwater W.	296/12	20	206/−10	18

Mean tilt-adjusted directions of maximum and intermediate axes at the four localities are shown in Table 7.

Hence principal axes of susceptibility are rotated clockwise in the same sense as the remanence directions. The declinations of the k_1 and k_2 axes at Freshwater West are ca. 40° more westerly than those at Freshwater East and somewhat more so than sites further to the east. The rotation of AMS is twice as much as rotation of the Variscan overprint (Table 3 and Fig. 7).

6. Discussion

This study reinforces the identification by earlier studies of a ubiquitous overprinting ChRM temporally linked to Variscan Orogenesis. A comparable magnetisation associated with compressional tectonics is also common along the foreland of the North American extension of this orogen. At the Appalachian margin it is attributed to growth of authigenic hematite mediated by fluid migration motivated by nappe stacking and lithospheric thickening (Oliver, 1986; McCabe and Elmore, 1989). In South Wales, remanence acquisition is contemporary with development of an Upper Carboniferous foreland basin (Setiabudidaya et al., 1994) and fluids responsible for chemical remagnetisation appear to have migrated up dip in sandstone aquifers of the Old Red Sandstone (Channell et al., 1992).

Comparative study of the three areas shows remagnetisation to be intimately linked with

orogeny and to be succeeded by late structural events. Results of Table 5 illustrate the value of separately untilting the limbs of folded structures. In these four examples the expected palaeofield direction is only recovered by untilting the southern limb more than the northern limb. It is, therefore, apparent that symmetrical fold structures, which had already been imparted to these rocks by the time of the magnetising event, have been steepened up from the south during succeeding stages of Variscan tectonism (during which they have probably also acquired their plunges).

The general clockwise rotation of the remanence vector to the west in South Wales, noted in Pembrokeshire by McClelland Brown (1983, see also Fig. 7) is confirmed by these tests but applies only to a total Variscan overprint. The largest rotation however, is concentrated within the orogen in the zone between Freshwater East and Freshwater West (Figs. 2 and 7, and Table 5). These locations are part of the same thrust slice and across this zone vergence directions are constant NNE-directed (Fig. 2). Hence it is inferred that differential rotation here cannot be a real structural effect. Instead we note: (1) the general increase in AMS between sites 10 and 11; (2) the larger rotation of AMS than remanence; and (3) the link between AMS and the fold structures (Sect. 5). Although no specific studies of strain have been made, it has been shown elsewhere (Stamatakos, 1991) that remanence in redbeds can be rotated by the effects of penetrative deformation with the least rotated vectors coming from sites with the lowest internal strain.

In contrast, the smaller but more regional rotation of the overprinted remanence shown in Fig. 7 is identified in unstrained, or only weakly strained, rocks. It is apparently the signature of a rotational event reflected in Variscan structural trends (Hancock et al., 1983). The differences in thrust transport directions between west Pembrokeshire (NNE) and the orogen to the east (NW to NNW) suggest that the fold belt became pinned to the east as it propagated northwards and westwards (McClelland Brown, 1983); this in turn applied a late rotational couple responsible for clockwise rotation of the remagnetisation.

7. Acknowledgements

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